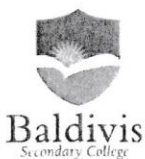


Name:			Date:	
Teacher:				
	Year 8 Mathematics		SCORE:	/50
	End of Term Test – Term 3 2019			47
	Topic – Equations, Data and Time <u>Full working out MUST be shown to get full marks for each question.</u>			
Total Time:	55 minutes			
Weighting:	6%			
Equipment:	To be provided by the student: Pens and Pencils, scientific calculator, A4 page of notes.			

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1. [17 marks]

For each of the following equations find the value of x .

a) $x + 4 = 12$

$x = 8$ ✓

b) $b - 15 = 62$

$b = 77$ ✓

c) $3x + 2 = -13$

$x = -5$ ✓✓

d) $3d = 63$

$d = 21$ ✓

e) $5e - 4 = 21$

$e = 5$ ✓✓

g) $3(x + 12) = 18$

$x = -6$ ✓✓

f) $3x = x + 4$

$2x = 4$ ✓
 $x = 2$ ✓

h) $4(l + 1) = 2(l - 3)$

$4l + 4 = 2l - 6$ ✓
 $2l = -10$ ✓
 $l = -5$ ✓

2. [10 marks]

Consider the following set of data

73, 57, 73, 62, 20, 69, 62, 73, 65, 67

a) Draw a stem and leaf plot for this data.

stem	leaf
2	0
3	
4	
5	7
6	2 2 5 7 9
7	3 3 3

key 7|3 = 73 ✓

b) Calculate;

I. Mean

$$\frac{621}{10} = 62.1 \quad \checkmark$$

II. Median

66 ✓

III. Mode

73 ✓

IV. Range

$$73 - 20 = 53 \quad \checkmark$$

b) Which score in the set of data is an outlier?

20 ✓

c) If the outlier is removed, what effect will this have on the;

Mean : 66.7 Increase ✓

and Mode : None ✓

3. [3 marks]

What type of data are the following?

Choose from categorical, discrete numerical or continuous numerical:

a the number of cans of Coke in a Coke machine ✓

discrete numerical.

b the favourite ice cream of each of the students in a class ✓

categorical.

c the length of time that a year 8 student can spend running ✓

continuous numerical.

4. [3 marks]

Leroy surveys his entire Year 8 class to find out the best holiday destination in Australia.

a) Is this a sample or a census? Why?

Sample. ✓✓

b) How is this survey biased?

Reason ✓

5. [5 marks] Complete the table

24-hour time	12-hour time
0017	12:17am
1440	2:40pm
1959	7:59pm
0001	12:01am
22 2243	10:43pm

6. [5 marks] Convert the following:

- a) 180 sec = 3 min ✓
- b) 3 years = 1095 days ✓
- c) 2 hours = 120 min ✓
- d) 4 weeks = 672 hours ✓✓

7. [2 marks]

Jarod went to the movies with some friends. If the movie started at 6:45pm and the movie went for 1 hour and 30 mins, at time did the movie finish?

- e) 12 hour time 8:15pm ✓
- f) 24 hour time 20:15 ✓

8. [2 marks]

A taxi departs at 18:52 from Sydney and arrives at 21:33 at Melbourne. How long did the taxi take for the trip?

2hr 41mins ✓✓

9. [3 marks]

Sarah needs to meet her friends at the cinema at 1:45pm. Her train ride will take 35 minutes, the bus ride to the station will take 15 minutes and Sarah will need to allow 10 minutes walking time at each end of the trip. At what time should Sarah leave home?

12:35pm ✓✓✓

----- End of Test -----